PAULEE GROUP · CENTER FOR QUANTUM TECHNOLOGY, KIST

Tutorial: Quantum Signal Processing

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Outline

1. **Signal processing, classically.**
2. **Quantum signal processing.**

- The QSP sequence
- The polynomial theorem with concrete (P, Q) examples
- The response examples.

3. **Fitting phases**

INTRO METHOD STRATEGY RESULTS ONGOING

JOURNAL CLUB The papers behind this tutorial

- J. M. Martyn, Z. M. Rossi, A. K. Tan, I. L. Chuang, *Grand Unification of Quantum Algorithms*, PRX Quantum **2**, 040203 (2021). A tutorial showing that search, phase estimation, and Hamiltonian simulation are all one algorithm, the quantum singular value transformation (QSVT). We use its QSP core: Sec. II A, the conventions of App. A, and the explicit phase lists of App. D.
- D. Motlagh, N. Wiebe, *Generalized Quantum Signal Processing*, PRX Quantum **5**, 020368 (2024).

01

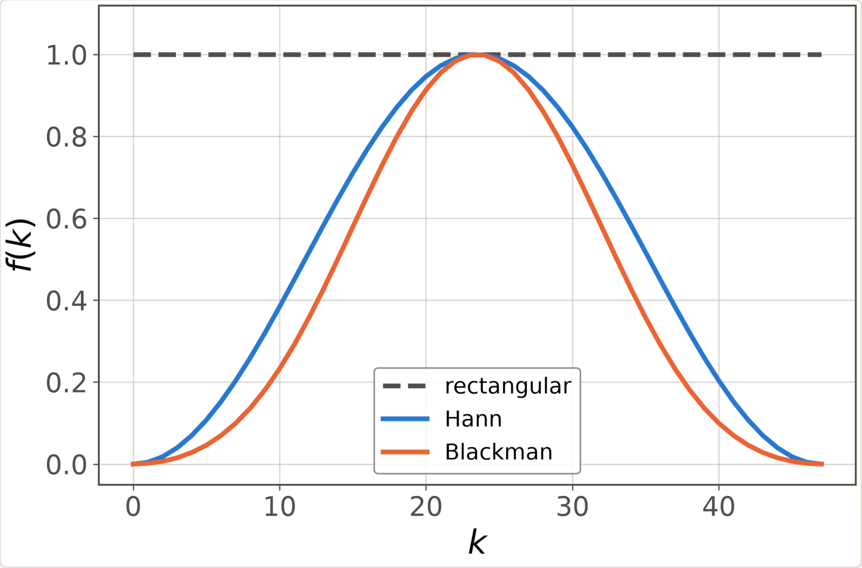
Signal processing, classically

From window coefficients to filters, including our own DDrf apodization

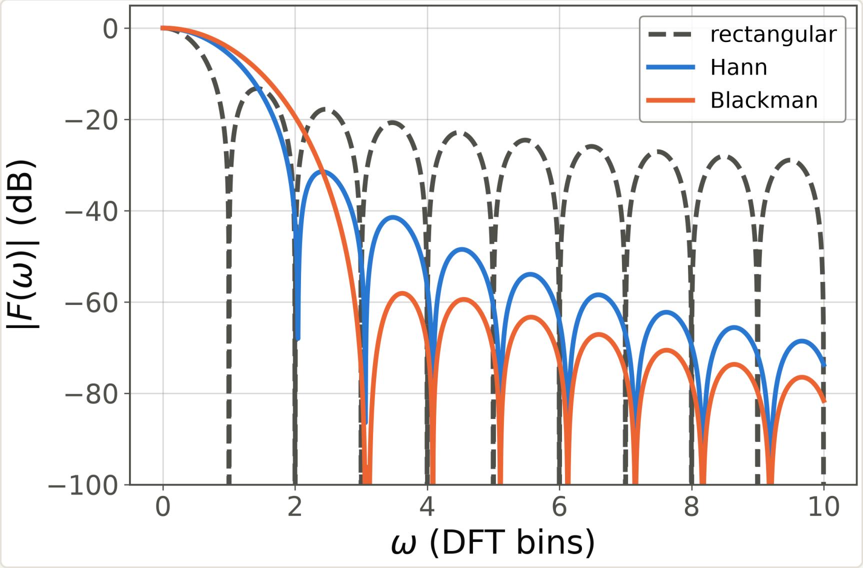
INTRO Windowing, also called apodization, is filter design

Multiplying data by a window $f(k)$ in time convolves the spectrum with the window's own Fourier transform, so the FFT of the taps **is** the filter:

$$\widetilde{S}(\omega) = (S * F)(\omega), \qquad F(\omega) = \sum_k f(k)\, e^{-i\omega k}.$$



Time domain: the chosen coefficients $f(k)$ (48 taps).



Frequency domain: rectangular sidelobes start at -13 dB; Blackman pushes them below -58 dB for a wider main lobe.

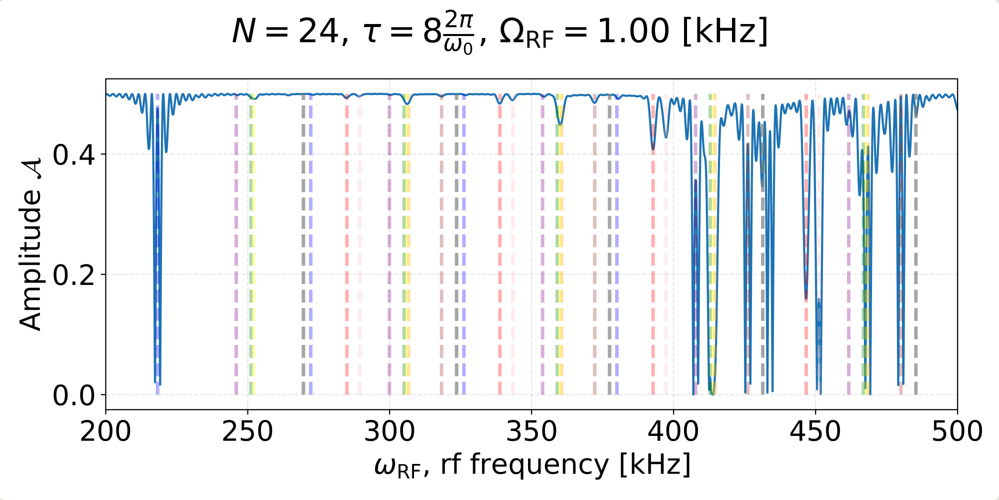
RESULTS

Recap: DDrf spectroscopy and its side peaks

DDrf drives a nuclear spin with N identical RF cells interleaved with electron π-pulses; sweeping ω_{RF} maps out resonances. In April we found **side peaks** persisting even where the gate should be unconditional; the detuned-Rabi formula explains envelope and lobe period,

$$\frac{1}{2}\,\mathrm{Tr}\,U_0 U_1^\dagger = 1 - \frac{2\Omega^2}{\Omega^2+\delta_1^2}\sin^2\!\left(\frac{\sqrt{\Omega^2+\delta_1^2}\,N\tau}{2}\right),$$

with δ_1 the detuning. These sidelobes are the Fourier structure of a **rectangular** drive envelope.

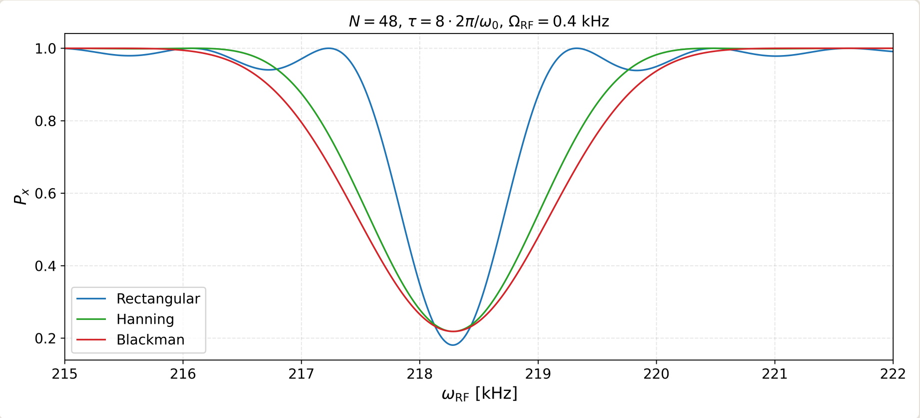


The observation from the April meeting: side peaks in the spectroscopy signal at parameters where the response should be flat. (source: lab meeting 2026-04-28)

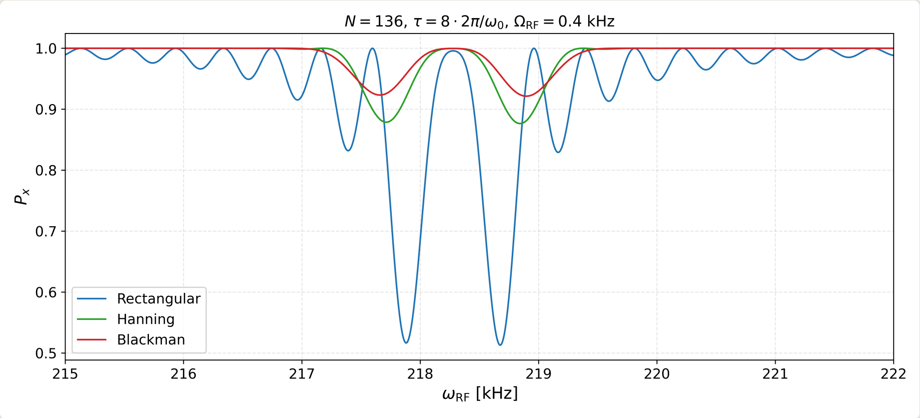
Suppression by apodized pulse envelopes

Replacing the constant amplitude by a windowed one, $\Omega_k = \Omega f(k)$, factorizes the normalized response into a sinc part and a kernel G fixed by the window,

$$\left|\frac{F(\delta_1)}{F(0)}\right|^2 = \mathrm{sinc}^2(u)\,|G(u)|^2, \qquad u = \frac{\delta_1}{2\Omega\bar{f}}.$$



Apodized envelopes flatten the detuned region, preserving the resonant peak ($N = 48$).



The suppression sharpens with the cell number ($N = 136$). (source: lab meeting 2026-04-28)

02

What is QSP doing?

Interleaved rotations realize polynomial response functions

METHOD The QSP sequence: one signal, many processing knobs

Two ingredients alternate [1, Sec. II A]. The **signal rotation** is a fixed x-rotation whose angle encodes the unknown signal $a \in [-1, 1]$,

$$W(a) = \begin{bmatrix} a & i\sqrt{1-a^2} \\ i\sqrt{1-a^2} & a \end{bmatrix}, \qquad a = \cos\frac{\theta}{2},$$

and it is applied d times, always the same. The **signal-processing rotations** $S(\phi_k) = e^{i\phi_k Z}$ are z-rotations through angles that we choose freely. The full sequence is

$$U_{\vec{\phi}} = e^{i\phi_0 Z} \prod_{k=1}^{d} W(a)\, e^{i\phi_k Z}, \qquad \vec{\phi} \in \mathbb{R}^{d+1}.$$

The structure is a composite pulse in which all the design freedom sits in the z-phases: the signal enters only through the repeated $W(a)$, and the $d+1$ knobs ϕ_k decide what the sequence does with it.

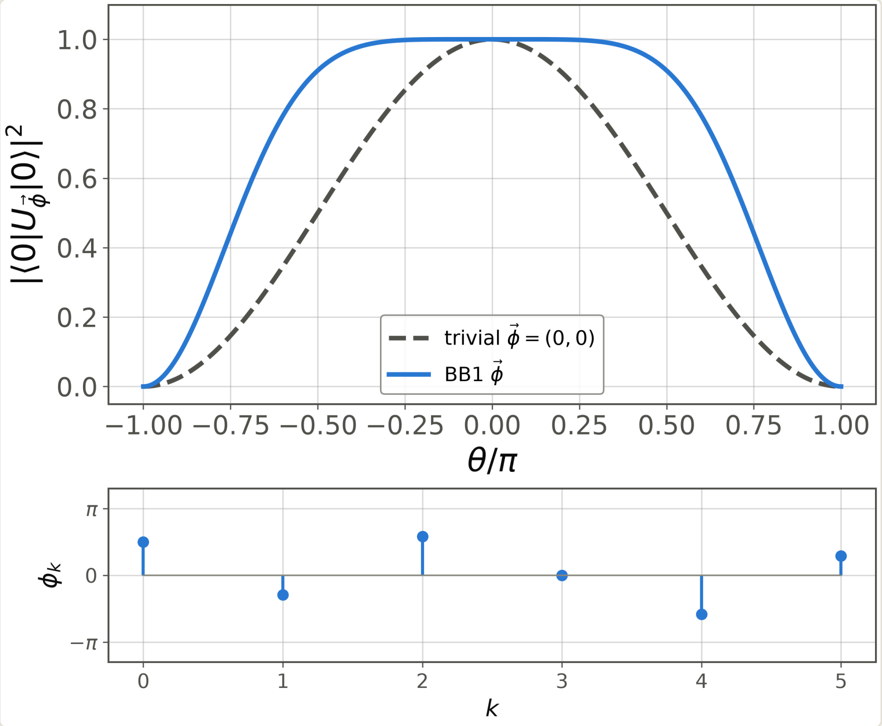
INTRO Composite pulses: signal processing on the Bloch sphere

This interleaved structure is older than quantum computing: it is an NMR **composite pulse**. There the rotation angle θ is set by the field, a miscalibrated pulse makes θ an unknown signal, and phased rotations are interleaved so that the transition probability responds to θ in a designed way.

BB1 uses five rotations with phases

$$\vec{\phi} = \left(\tfrac{\pi}{2}, -\eta, 2\eta, 0, -2\eta, \eta\right), \quad \eta = \tfrac{1}{2}\cos^{-1}\left(-\tfrac{1}{4}\right),$$

which flattens the response to $1 - \frac{5}{8}(\theta/2)^6$ near $\theta = 0$ and switches sharply near $\theta \approx 2\pi/3$.



Reproduction of Fig. 1 of [1] with the phase vector below: the bare rotation responds as $\cos^2(\theta/2)$, the BB1 sequence holds the qubit unflipped over a wide band.

METHOD

The QSP theorem: the response is a designed polynomial

Theorem 1 of [1]. The sequence $U_{\vec{\phi}}$ always takes the form

$$U_{\vec{\phi}} = \begin{bmatrix} P(a) & iQ(a)\sqrt{1-a^2} \\ iQ^*(a)\sqrt{1-a^2} & P^*(a) \end{bmatrix},$$

and conversely a $\vec{\phi}$ exists for **any** polynomials P, Q with (i) $\deg P \le d$, $\deg Q \le d-1$, (ii) parity $d \bmod 2$ and $(d-1) \bmod 2$, (iii) $|P|^2 + (1-a^2)|Q|^2 = 1$.

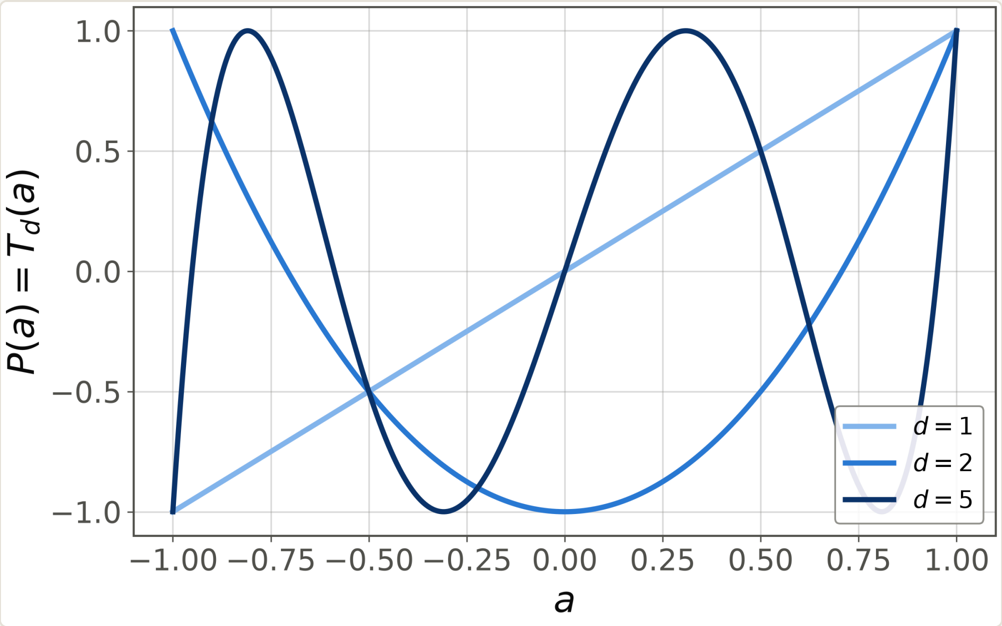
The forward direction is an induction: each extra $W(a)\, e^{i\phi_k Z}$ raises the degree by one and flips the parities, while unitarity enforces condition (iii). The converse is the useful direction: the reachable response family is known **exactly**, not perturbatively. The next slide makes (P, Q) concrete.

METHOD What are P and Q, concretely?

Take no processing, $\vec{\phi} = (0, \ldots, 0)$: the sequence is bare repetition, $U = W(a)^d$.

- $d = 1$: $P(a) = a$, $Q(a) = 1$.
- $d = 2$: squaring $W(a)$ gives $P(a) = 2a^2 - 1, Q(a) = 2a$.
- In general $P = T_d(a)$, $Q = U_{d-1}(a)$, Chebyshev of the first and second kind; condition (iii) is the identity $T_d^2 + (1 - a^2)\, U_{d-1}^2 = 1$.

The phases ϕ_k reshape this family into any other admissible pair.



$P(a) = T_d(a)$ at trivial phases for $d = 1, 2, 5$, from our evaluator (matches to 10^{-12}).

METHOD Which functions can the response be?

P **alone is boxed in at the endpoints.** At $a = \pm 1$ the signal rotation is trivial, $W(\pm 1) = \pm I$, so the whole sequence collapses to a single z-rotation and condition (iii) forces $|P(\pm 1)| = 1$. A modest target like $0.5\,\mathrm{sign}(a)$ is therefore impossible as P itself.

The fix is to read only the real part in the $|+\rangle$ **basis** [1]:

$$\langle +|U_{\vec{\phi}}|+\rangle = \mathrm{Re}\,P(a) + i\,\mathrm{Re}\,Q(a)\sqrt{1-a^2}.$$

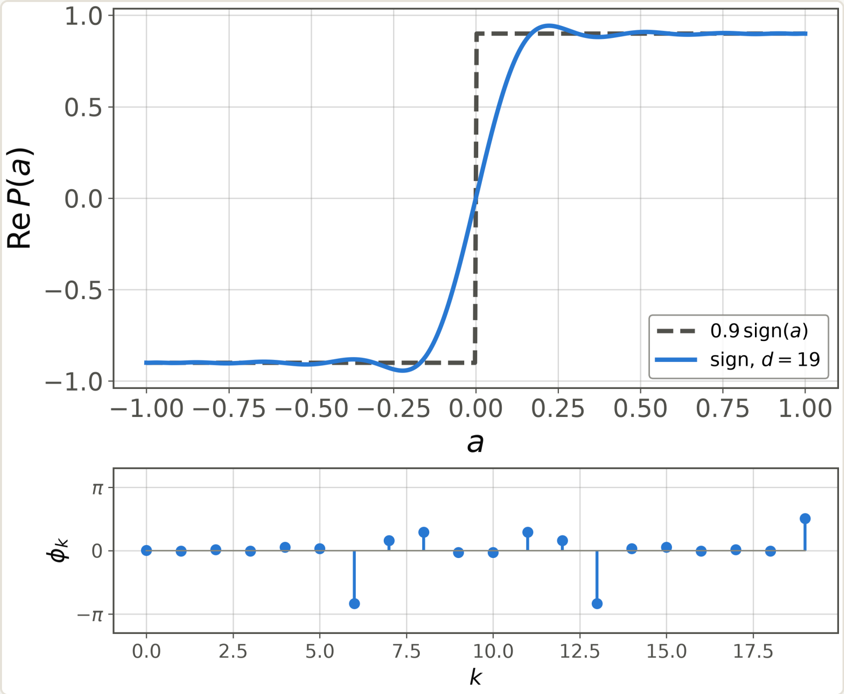
The target is encoded in $\mathrm{Re}\,P$, while $\mathrm{Im}\,P$ absorbs the unit-modulus burden: the published sign list has $\mathrm{Re}\,P(1) = 0.90$ with $|P(1)| = 1$ exactly, the rest being imaginary.

Result. $\mathrm{Re}\,P$ can be **any** real polynomial with parity $d \bmod 2$ and $|\mathrm{Poly}(a)| \le 1$. This is the design space used by the gallery and by our fits.

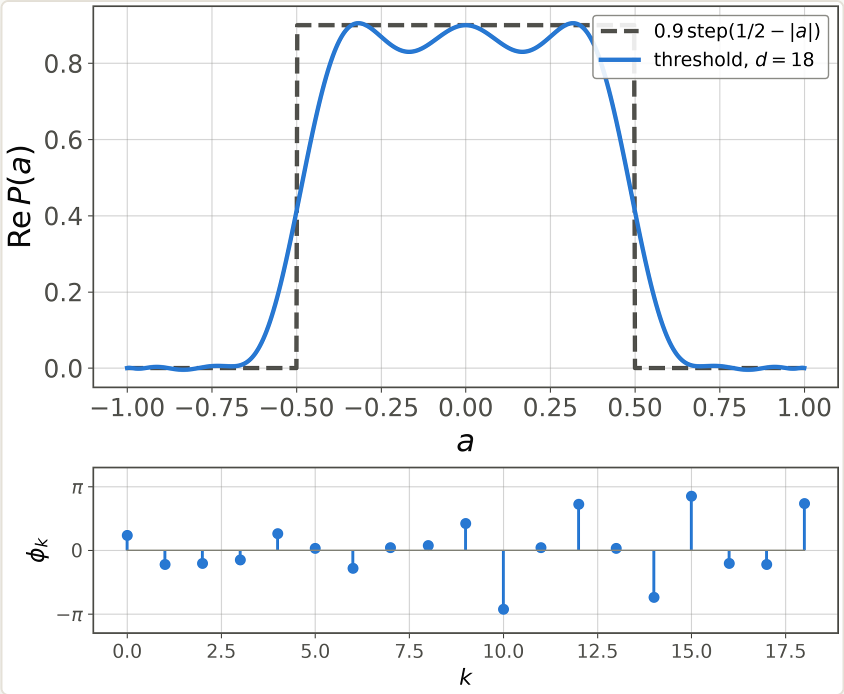
Watch the convention: papers and libraries differ by $\pm\pi/4$ phase shifts and signal rescalings [1, App. A]; a live example follows on the trigonometric gallery slide.

METHOD

What can we do? Polynomial design is algorithm design.



Sign function ($d = 19$): the decision primitive behind amplitude amplification and search.

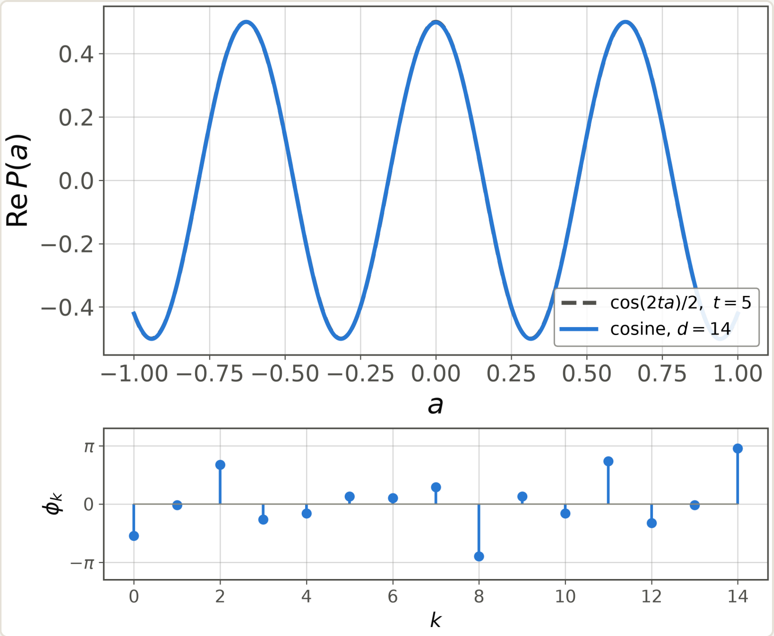


Threshold function ($d = 18$): a band-pass response, used for eigenvalue thresholding and eigenstate filtering.

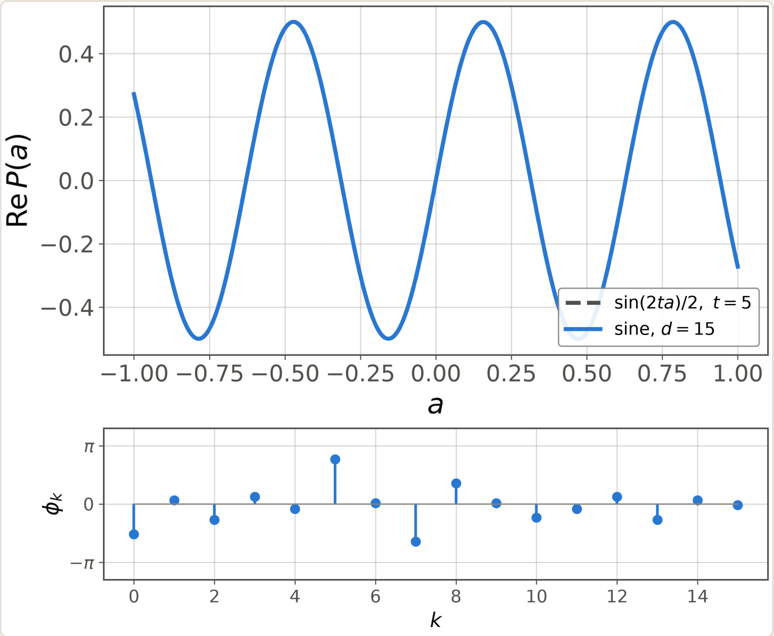
Appendix D of [1] also lists phases for $\cos$, $\sin$ (next slide), $1/a$, $e^{-\beta a}$, and ReLU; each response is drawn with its phase vector $\vec{\phi}$, the program that produces it.

METHOD

Gallery: cosine and sine for Hamiltonian simulation



Cosine list [1, App. D4], $t = 5, d = 14.$



Sine list [1, App. D4], $t = 5, d = 15.$

$$\vec{\phi}_{\cos} = (-1.71, -0.05, 2.12, -0.83, -0.50, 0.41, 0.33, 0.91, -2.81, 0.41, -0.50, 2.31, -1.02, -0.05, 3.00)$$
$$\vec{\phi}_{\sin} = (-1.63, 0.21, -0.84, 0.40, -0.27, 2.41, 0.05, -2.03, 1.11, 0.05, -0.73, -0.27, 0.40, -0.84, 0.21, -0.06)$$

Hamiltonian simulation uses the pair as $e^{-i\mathcal{H}t} = \cos(\mathcal{H}t) - i\sin(\mathcal{H}t)$; these two angle lists are the entire program (rendered as $\cos(2ta)/2$, $\sin(2ta)/2$ in our convention [1, App. A]).

METHOD Where do the phases come from?

Theorem 1 is existence; the numbers take an algorithm, as implemented in **pyqsp** (Chuang group; generated the App.-D lists) and **QSPPACK** (Lin group, MATLAB):

- **The target polynomial comes first**: a bounded polynomial approximant of the desired function, via erf and Taylor constructions, Chebyshev interpolation, or Remez exchange [1, App. D].
- **Exact factorization**: root-finding on the complementary polynomial (GSLW), or Laurent-polynomial division at machine precision (Haah; Chao *et al.*; `--method laurent`). Numerically delicate at large degree.
- **Iteration and optimization**: quasi-Newton on symmetric phase vectors (Dong *et al.*, QSPPACK; `--method sym_qsp`), fast and stable with degrees in the thousands routine; plain gradient descent on the response error also works (`--method tf`).

Our fits in Section 3 are the minimal version of the last route: same model, exact gradient.

ONGOING Where this goes: from one number to a whole matrix

So far one qubit encoded one number a. **Qubitization** embeds a Hamiltonian $\mathcal{H} = \sum_\lambda \lambda|\lambda\rangle\langle\lambda|$ as a block of a unitary,

$$U = \begin{bmatrix} \mathcal{H} & \sqrt{1-\mathcal{H}^2} \\ \sqrt{1-\mathcal{H}^2} & -\mathcal{H} \end{bmatrix},$$

and each eigenvalue λ then lives in its own private Bloch sphere with signal $a = \lambda$. The **same** phase sequence acts on all of them at once [1, Sec. II C]:

$$\mathrm{Poly}(\mathcal{H}) = \sum_\lambda \mathrm{Poly}(\lambda)\,|\lambda\rangle\langle\lambda|.$$

With the Appendix-D polynomials this one template becomes Grover search (sign), phase estimation (threshold), Hamiltonian simulation (cosine, sine), and matrix inversion ($1/a$): the "grand unification" of the title. Today we stop at the single-qubit layer.

03

Fitting phases by our own hand

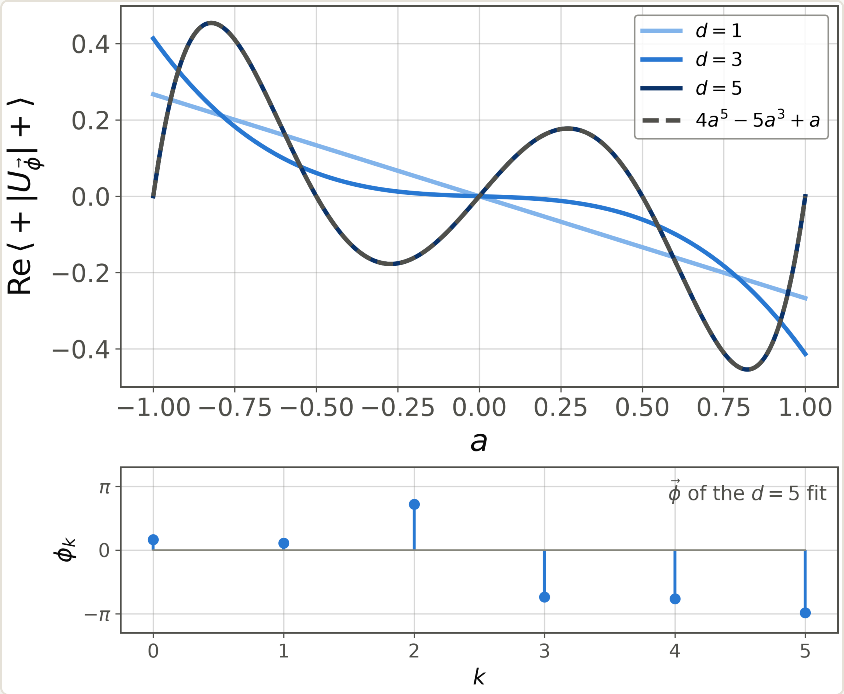
A numpy evaluator, validated against the paper, then optimized

METHOD Setup and validation gates

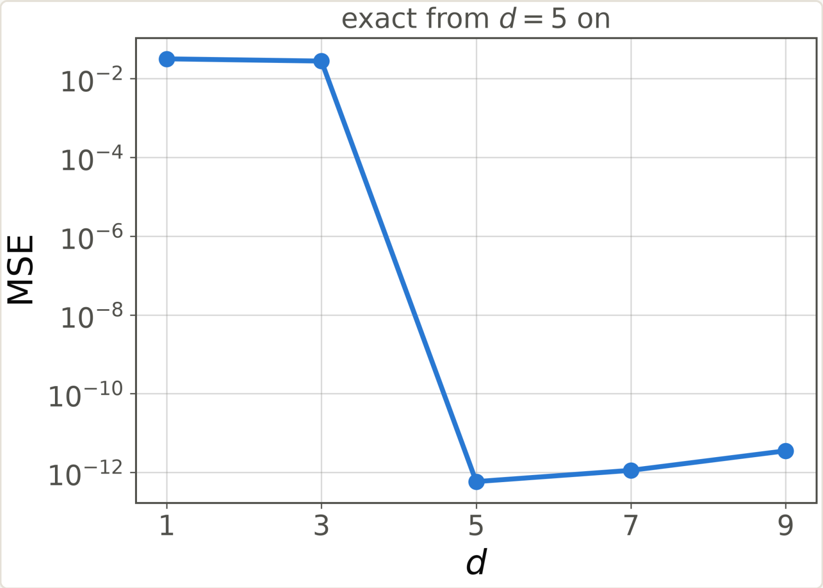
- **Model.** We implement $U_{\vec{\phi}}(a)$ as the literal Eq.-(3) product of 2×2 matrices and read out the response $\mathrm{Re}\,\langle + | U_{\vec{\phi}} | + \rangle$. Fitting minimizes the mean squared error to a target on a grid of 51 points in $a \in [-1, 1]$.
- **Optimizer.** L-BFGS with the exact gradient (obtained by differentiating the product chain, $\partial U / \partial \phi_k = P_k\,(iZ)\,Q_k$ with prefix and suffix products P_k, Q_k), deterministic multi-start with seed 42.
- **Validation before any fitting.** Trivial phases reproduce $T_d(a)$ below 10^{-12}; BB1 matches its closed-form response to 10^{-15}; the published $d = 19$ sign list [1, App. D2] reproduces Fig. 21 of the paper, odd to machine precision with plateau values in $[0.881, 0.936]$.
- **Reference point.** The PennyLane demo *Function fitting using QSP* trains the same model with PyTorch SGD for up to $25{,}000$ iterations; with the exact gradient, every fit on the next slides converges in less than three seconds.

RESULTS

Warm-up target: exact once d reaches the degree



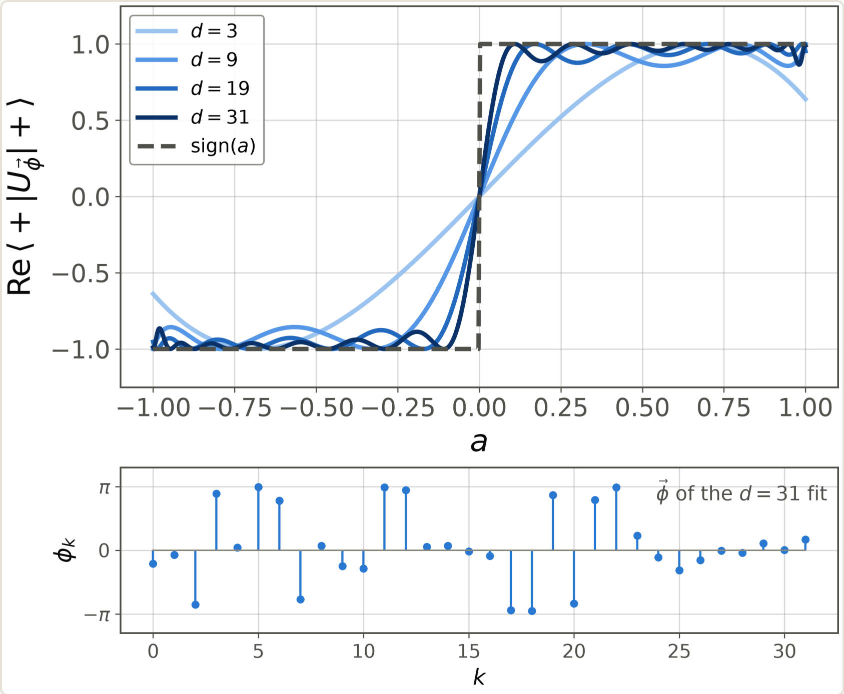
Fits at $d = 1, 3, 5$ with the phases of the exact $d = 5$ fit; the dashed target lies on the $d = 5$ curve.



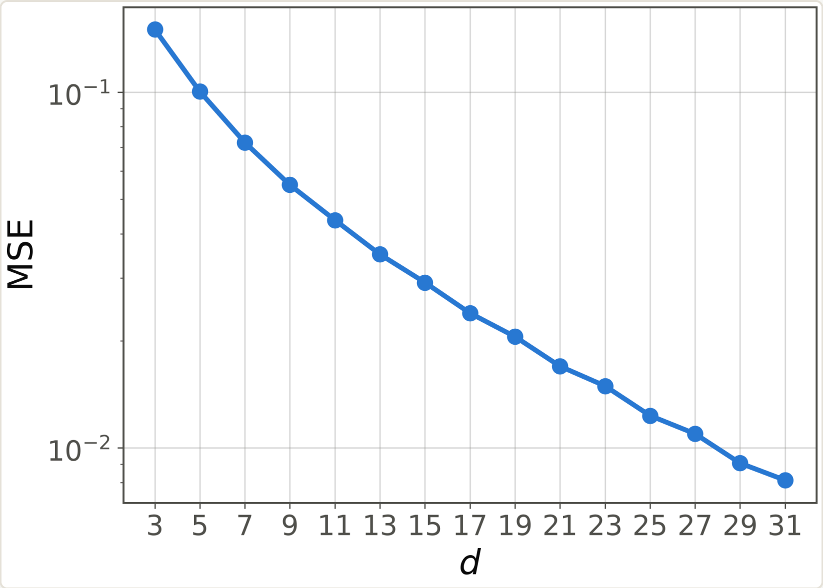
The error collapses at $d = 5$, from 2.8×10^{-2} to 5.9×10^{-13}, then stays at machine precision.

The response of the PennyLane demo target $4a^5 - 5a^3 + a$ improves with d and becomes **exact** once the sequence can carry it: a degree-5 odd polynomial needs exactly $d = 5$.

RESULTS Step target: systematic sharpening with d



Fits of $\mathrm{sign}(a)$ *at* $d = 3, 9, 19, 31$ *with the phases of the* $d = 31$ *fit; the plateaus hug* ± 1 *with ripples.*

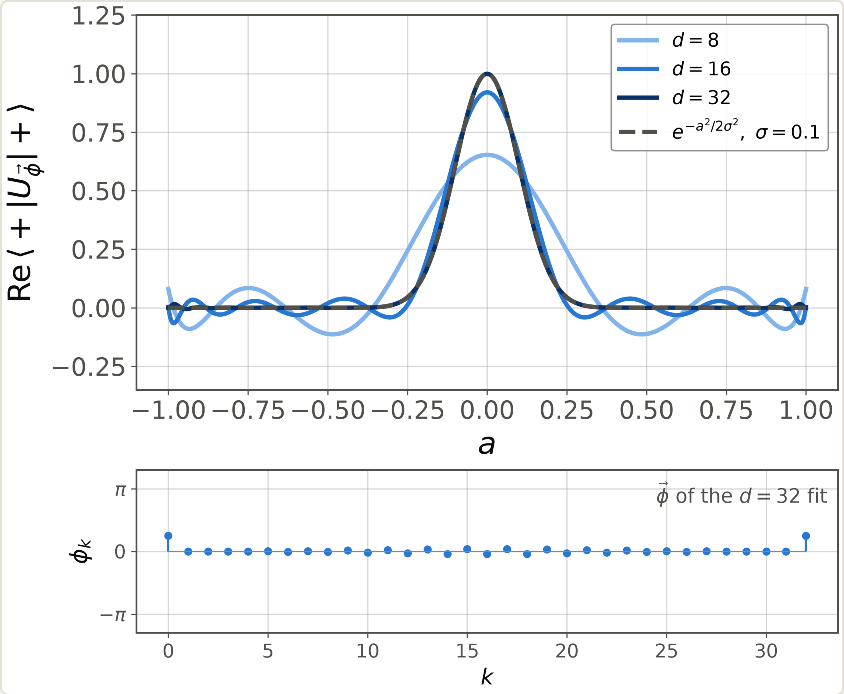


MSE decreases monotonically, 1.5×10^{-1} *at* $d = 3$ *to* 8.1×10^{-3} *at* $d = 31$*; no finite degree is exact for a discontinuity.*

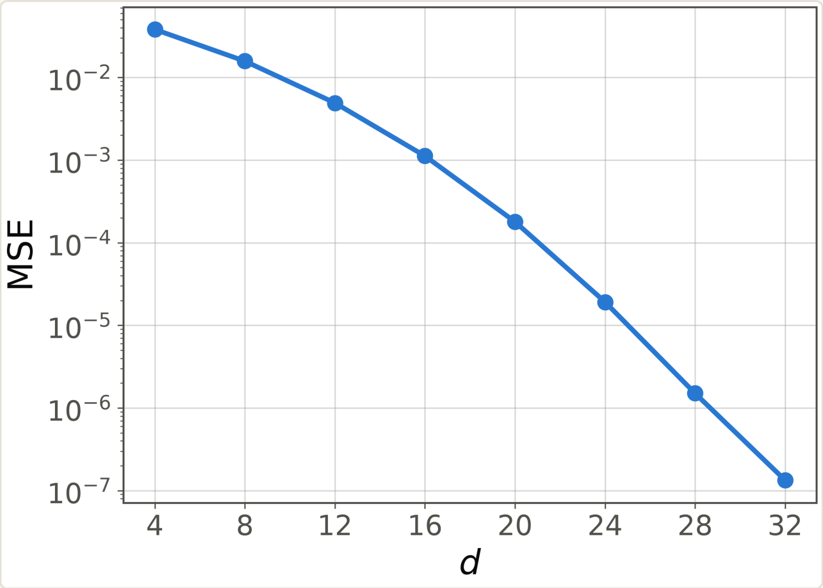
Sharper transition, lower ripple, higher degree: the window-design trade in polynomial language. The paper instead smooths the target to $\mathrm{erf}(ka)$ [1, App. D2].

RESULTS

Delta-function target: resolution costs degree



Fits of a delta-like spike $e^{-a^2/2\sigma^2}$, $\sigma = 0.1$, at $d = 8, 16, 32$, with the phases of the $d = 32$ fit.



MSE falls from 3.8×10^{-2} at $d = 4$ to 1.3×10^{-7} at $d = 32$.

A spike of width σ needs $d \sim 1/\sigma$: resolution is bought with sequence length, as the DDrf line width shrinks with N. This response is eigenstate filtering [1, App. D8].

ONGOING

Summary

QSP is response engineering promoted to an exact design principle, and we have used its classical shadow before.

- The sequence interleaves one fixed signal rotation $W(a)$ with chosen z-phases; the response is exactly a bounded polynomial of the signal, and every admissible polynomial is reachable (Theorem 1 of [1]).
- Polynomial design is algorithm design: sign, threshold, cosine, and $1/a$ responses become search, filtering, simulation, and inversion once the signal is an eigenvalue (QSVT).
- Our DDrf apodization solved the same shape of problem: per-cell amplitudes shape the response over detuning, and in both worlds more repetitions buy sharper features.
- Fitting phases is easy at tutorial scale: a numpy script validates against the published lists and fits polynomial, step, and delta targets in seconds, improving systematically with d. A possible follow-up: QSP-style exact synthesis for our pulse-envelope design.

References

1. J. M. Martyn, Z. M. Rossi, A. K. Tan, I. L. Chuang, *Grand Unification of Quantum Algorithms*. PRX Quantum **2**, 040203 (2021).
2. D. Motlagh, N. Wiebe, *Generalized Quantum Signal Processing*. PRX Quantum **5**, 020368 (2024).
3. G. H. Low, T. J. Yoder, I. L. Chuang, *Methodology of Resonant Equiangular Composite Quantum Gates*. Phys. Rev. X **6**, 041067 (2016).
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6. QSPPACK, `github.com/qsppack/QSPPACK` (phase-factor solvers; Y. Dong, X. Meng, K. B. Whaley, L. Lin, Phys. Rev. A **103**, 042419 (2021)).
7. pyqsp, `github.com/ichuang/pyqsp` (source of the Appendix-D phase lists of [1]; Laurent method of R. Chao *et al.*, `arXiv:2003.02831`).
8. PennyLane demo, *Function fitting using quantum signal processing*, `pennylane.ai/demos/function_fitting_qsp`.

Thank you

Questions & discussion

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